

```
1 using PlutoUI
320 ms

1 using PlutoSlides
29.8 ms

1 using CairoMakie
7.8 se

1 import WGLMakie
1.2 se

1 using LaTeXStrings
1.4 ms

1 using Distributions
1.8 ms

1 slide_mode_settings(footer_left="David Métivier", footer_center=md"The Riemann-Lebesgue lemma", font_size=fontsize, color_subtitle_bg =
"#9558B2")
453 ms

fontsize = 22

1 fontsize = 22
41.2 μs

Slide Mode

1 slide_mode_button(start_in_slide_mode_html=true, start_slide = 1)
```

The Riemann–Lebesgue lemma

or why marathoners running in circles end up evenly spread

David Métivier

The logo for INRAE, featuring the letters 'INRAE' in a bold, teal, sans-serif font. The letter 'A' is stylized with a leaf-like shape integrated into its right side.

Made with [Julia](#), [Pluto.jl](#), [PlutoSlides.jl](#), and [Makie.jl](#).

The lemma

Statement

Lemma (Riemann–Lebesgue)

Let $f \in L^1(\mathbb{R})$ and define its Fourier transform

$$\hat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx, \quad \xi \in \mathbb{R}.$$

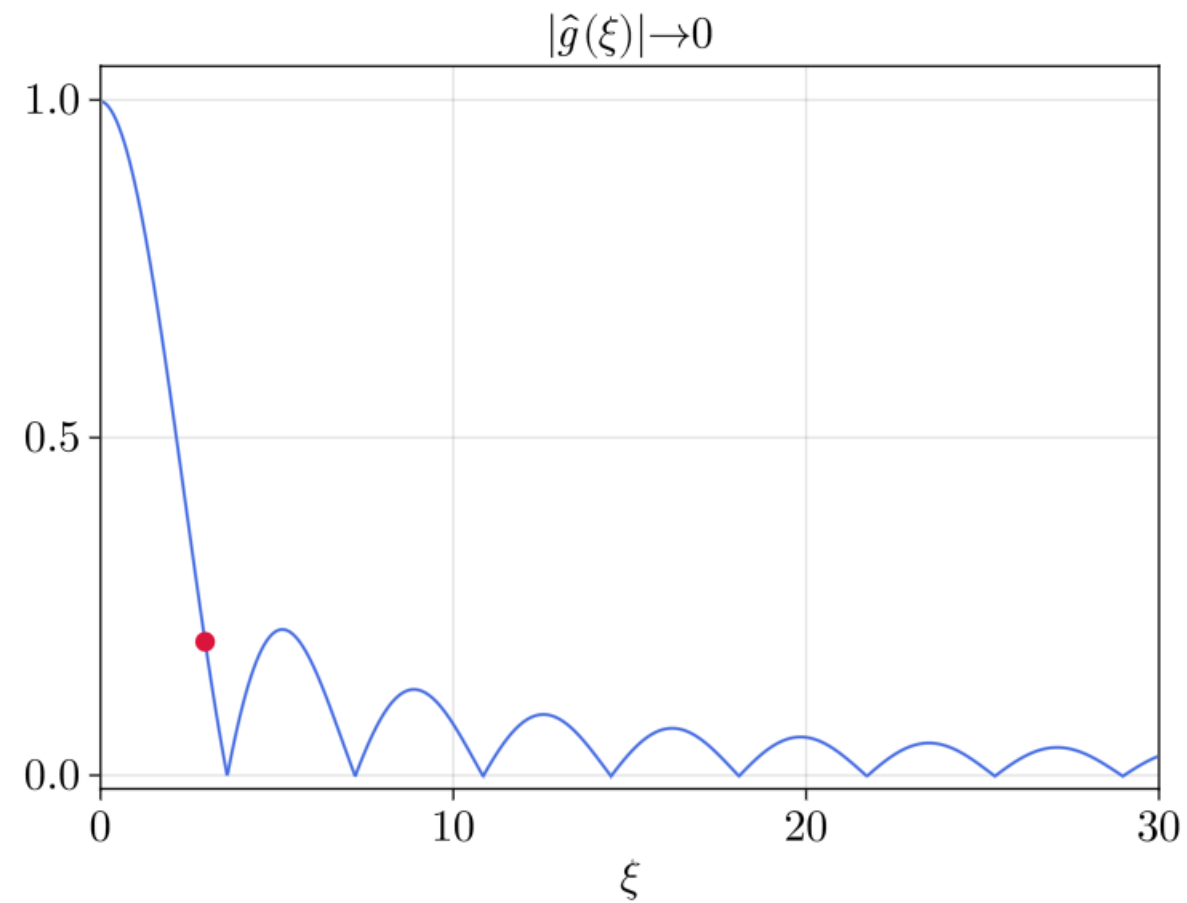
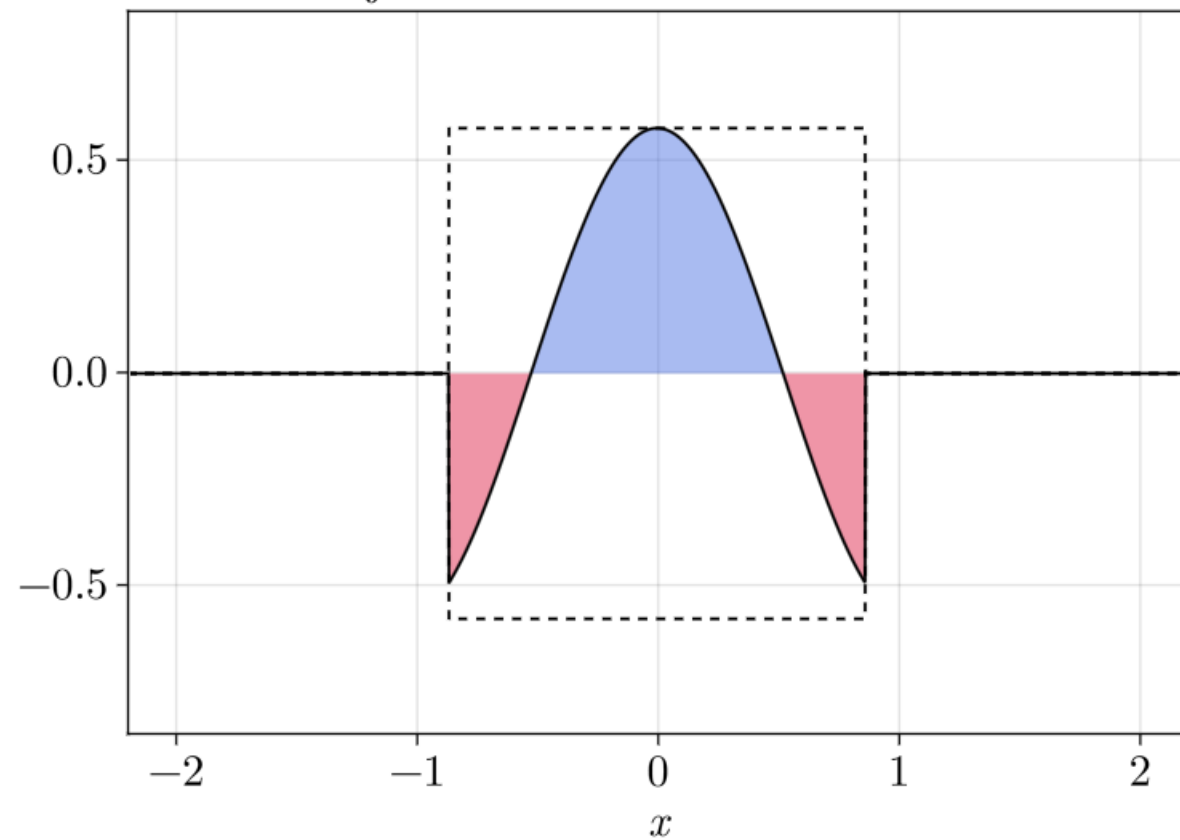
Then $\hat{f}$ is continuous and **vanishes at infinity**: $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.

- Same statement on $\mathbb{R}^n$, and on the circle: Fourier coefficients $\hat{f}(k) \rightarrow 0$ as $|k| \rightarrow \infty$.
- Intuition: against a fixed integrable f , the fast oscillations $e^{-i\xi x}$ **cancel out**.
- No rate in general: $\hat{f}$ can go to 0 arbitrarily slowly. A **rate** needs **regularity** of f .

Intuition: Oscillations cancel out

Take for f a probability density g Uniform (jump) ▾ frequency $\xi =$ 3.0

$$\int g(x) \cos(\xi x) dx = \hat{g}(\xi) = 0.199$$



Proof

Step 1 — indicators. For $f = \mathbf{1}_{[a,b]}$,

$$\hat{f}(\xi) = \int_a^b e^{-i\xi x} dx = \frac{e^{-i\xi a} - e^{-i\xi b}}{i\xi}, \quad \text{so} \quad |\hat{f}(\xi)| \leq \frac{2}{|\xi|} \xrightarrow{|\xi| \rightarrow \infty} 0.$$

Step 2 — step functions. By linearity, the same holds for every step function $\varphi = \sum_{j=1}^m c_j \mathbf{1}_{[a_j, b_j]}$.

Step 3 — density. Step functions are dense in $L^1(\mathbb{R})$: for $\varepsilon > 0$, pick φ with $\|f - \varphi\|_{L^1} < \varepsilon$. Since $|\hat{f}(\xi) - \hat{\varphi}(\xi)| \leq \|f - \varphi\|_{L^1}$ for every ξ ,

$$\limsup_{|\xi| \rightarrow \infty} |\hat{f}(\xi)| \leq \varepsilon + \lim_{|\xi| \rightarrow \infty} |\hat{\varphi}(\xi)| = \varepsilon.$$

As ε is arbitrary, $\hat{f}(\xi) \rightarrow 0$. ■

A one-line proof

Shifting by half a period flips the sign of the wave, $e^{-i\xi(x+\pi/\xi)} = -e^{-i\xi x}$, hence

$$\hat{f}(\xi) = - \int_{\mathbb{R}} f\left(x + \frac{\pi}{\xi}\right) e^{-i\xi x} dx \implies |\hat{f}(\xi)| \leq \frac{1}{2} \|f - f(\cdot + \frac{\pi}{\xi})\|_{L^1} \xrightarrow{|\xi| \rightarrow \infty} 0$$

by continuity of translations in L^1 . ■

Bonus: a rate for free. If f has bounded variation, $\|f - f(\cdot + h)\|_{L^1} \leq |h| \text{TV}(f)$, so

$$|\hat{f}(\xi)| \leq \frac{\pi \text{TV}(f)}{2 |\xi|}.$$

Regularity $\implies$ decay

Integrating by parts k times (if $f, f', \dots, f^{(k)} \in L^1$):

$$\widehat{f^{(k)}}(\xi) = (i\xi)^k \hat{f}(\xi) \implies |\hat{f}(\xi)| = \frac{|\widehat{f^{(k)}}(\xi)|}{|\xi|^k} = o(|\xi|^{-k}) \quad (\text{Riemann-Lebesgue for } f^{(k)}).$$

A jump in $f^{(k)}$ gives exactly $|\hat{f}(\xi)| \asymp |\xi|^{-(k+1)}$. Our test densities, all with standard deviation 0.5:

$g(x)$ on its support	regularity	decay of $\hat{g}(\xi)$
Uniform $\propto 1$	jump	ξ^{-1}
Epanechnikov $\propto 1 - x^2$	C^0 , kink	ξ^{-2}
Biweight $\propto (1 - x^2)^2$	C^1	ξ^{-3}
Triweight $\propto (1 - x^2)^3$	C^2	ξ^{-4}
Gaussian $\propto e^{-x^2/2\sigma^2}$	analytic	$e^{-\sigma^2\xi^2/2}$

Fun fact: $(1 - x^2)^k$ on $[-1, 1]$ is the density (up to a constant) of the median of $2k + 1$ independent uniform variables on $[-1, 1]$.

Example: the marathon

The model

N runners on a circular track $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, all starting at $\theta = 0$. Runner j keeps a constant speed v_j , drawn independently from a density g (mean 4.0, standard deviation 0.5):

$$\theta_j(t) = v_j t \mod 2\pi.$$

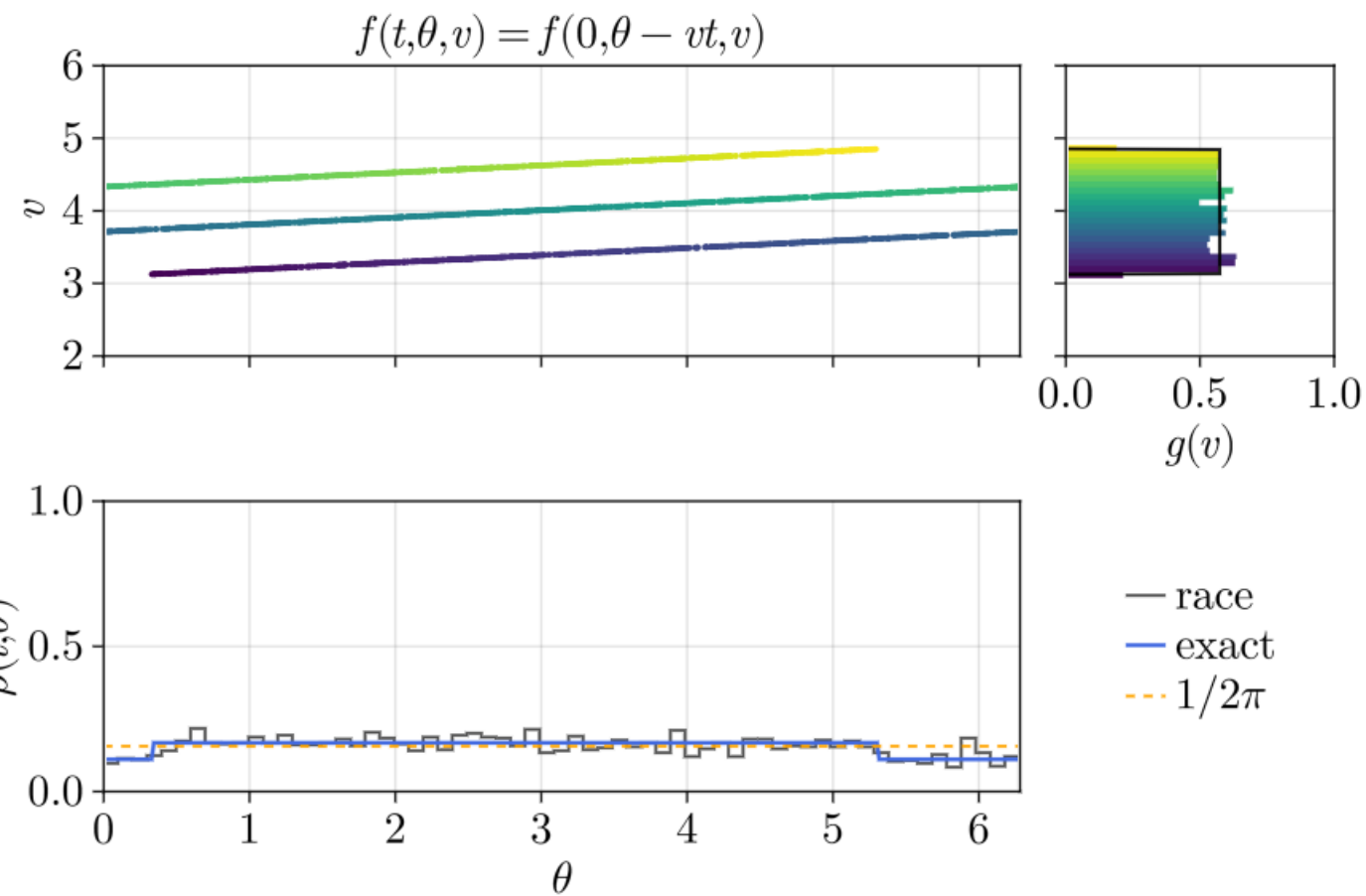
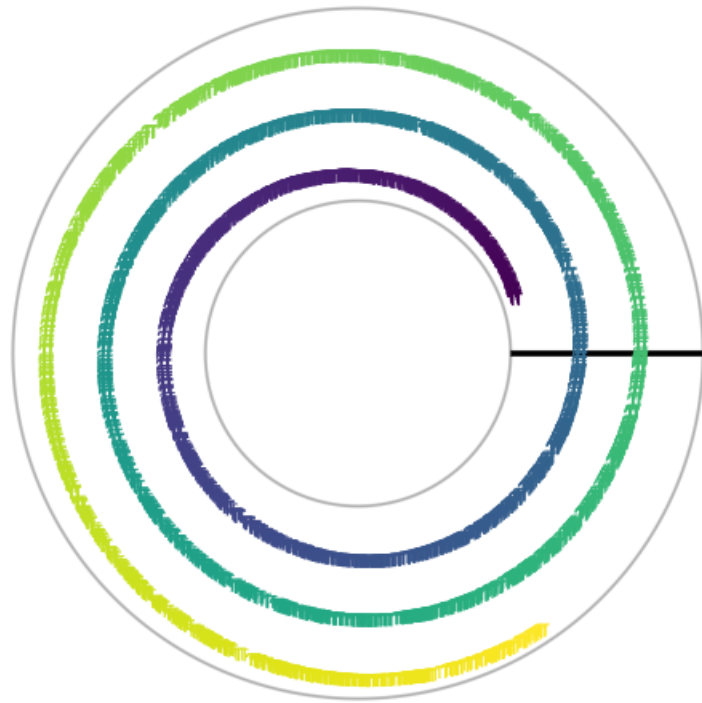
Kinetic view. The density $f(t, \theta, v)$ of runners at position θ with speed v solves the free transport (advection) equation

$$\partial_t f + v \partial_\theta f = 0, \quad f(0, \theta, v) = \delta_0(\theta) g(v).$$

For each speed, the solution is a **traveling wave**: $f(t, \theta, v) = f(0, \theta - vt, v)$. Runners with the same speed stay together forever...

1. What does happen to $f(t, \theta, v)$?
2. Does the crowd $\rho(t, \theta) = \int f(t, \theta, v) dv$ spread evenly around the track? And **how fast**?

Let's race!

Speed law Uniform (jump) $N =$ 3000 exact ρ ☒Time $t =$ 10.13 $t = 10.13$ $|c_1(t)| = 0.074$ 

Uniform law: the slowest and fastest runners keep a **sharp edge** in ρ that only shrinks like $1/t$. Gaussian law: ρ is flat within a few laps. Meanwhile $g(v)$, the other marginal, never moves.

Mixing is Riemann–Lebesgue

$$\partial_t f + v \partial_\theta f = 0, \quad f(0, \theta, v) = \delta_0(\theta) g(v), \quad \rho(t, \theta) = \int f(t, \theta, v) dv$$

Fourier coefficients of the crowd on the circle:

$$c_k(t) = \int_{\mathbb{T}} \rho(t, \theta) e^{-ik\theta} d\theta = \mathbb{E}[e^{-ikvt}] = \int_{\mathbb{R}} g(v) e^{-i(kt)v} dv = \hat{g}(kt).$$

Hence $\rho(t, \theta) = \frac{1}{2\pi} \left(1 + \sum_{k \neq 0} \hat{g}(kt) e^{ik\theta} \right)$, and by **Riemann–Lebesgue** every mode dies: $\rho(t, \cdot) \rightharpoonup \frac{1}{2\pi}$. The crowd spreads out evenly.

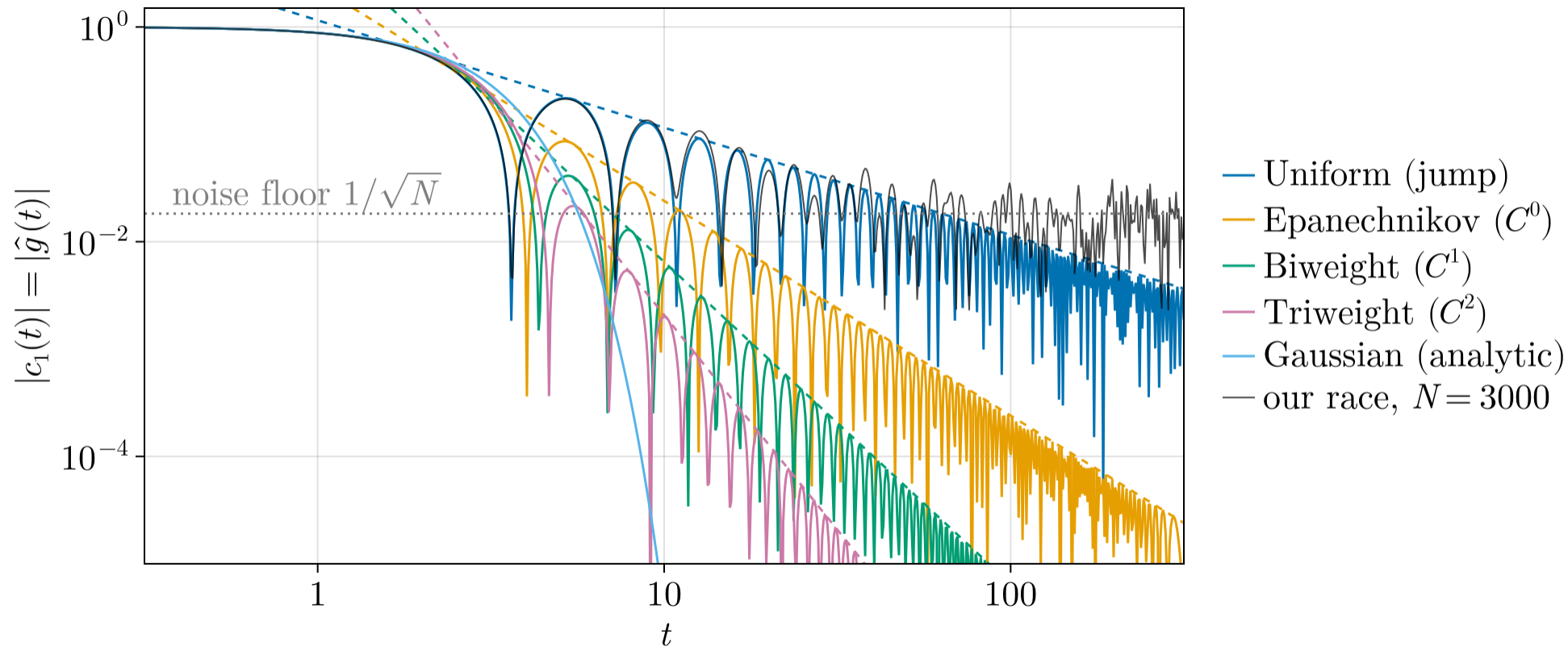
The mixing rate is the decay rate of $\hat{g}$, i.e. the regularity of the speed distribution: jump $\Rightarrow t^{-1}$, discontinuous first derivative $\Rightarrow t^{-2}$, ..., Gaussian $\Rightarrow e^{-\sigma^2 t^2/2}$.

The kinetic density f itself never relaxes: it filaments in phase space and converges only weakly. This *phase mixing* is the mechanism behind Landau damping (Mouhot & Villani, 2011) a damping without dissipation (energy is conserved).

How fast? Theory vs. the race

Show

☒ Uniform (jump)
 ☒ Epanechnikov (C^0)
 ☒ Biweight (C^1)
 ☒ Triweight (C^2)
 ☒ Gaussian (analytic)



Solid: $|\hat{g}(t)|$, exact (cf from Distributions.jl). Dashed: its asymptotic envelope $C t^{-k}$. Black: $\left| \frac{1}{N} \sum_j e^{i\theta_j(t)} \right|$ measured on the race above.

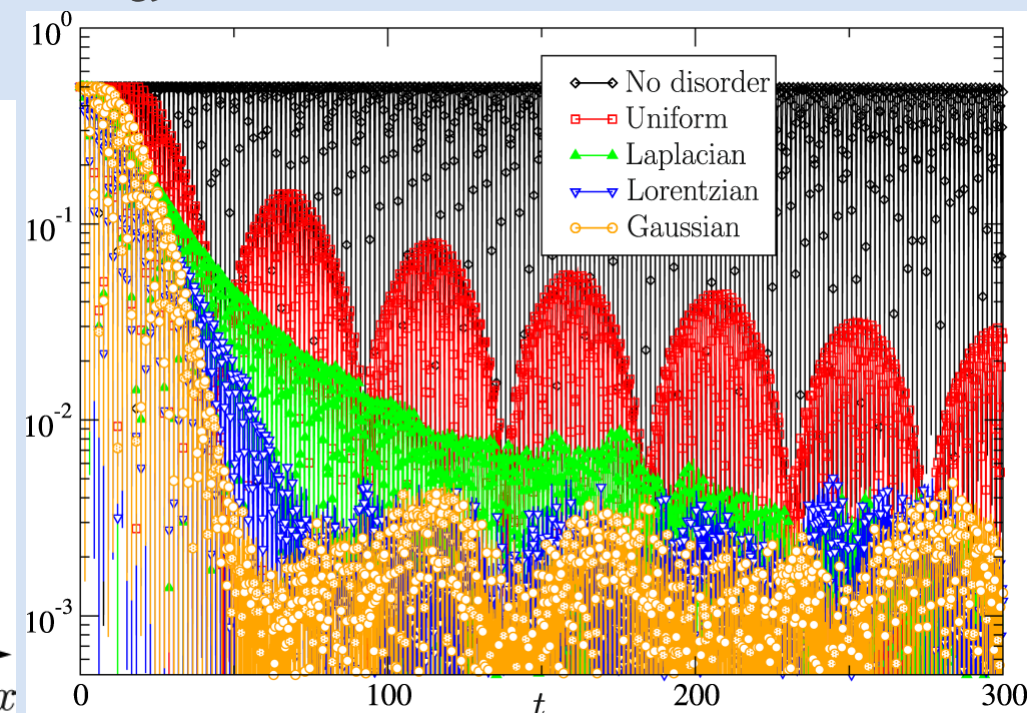
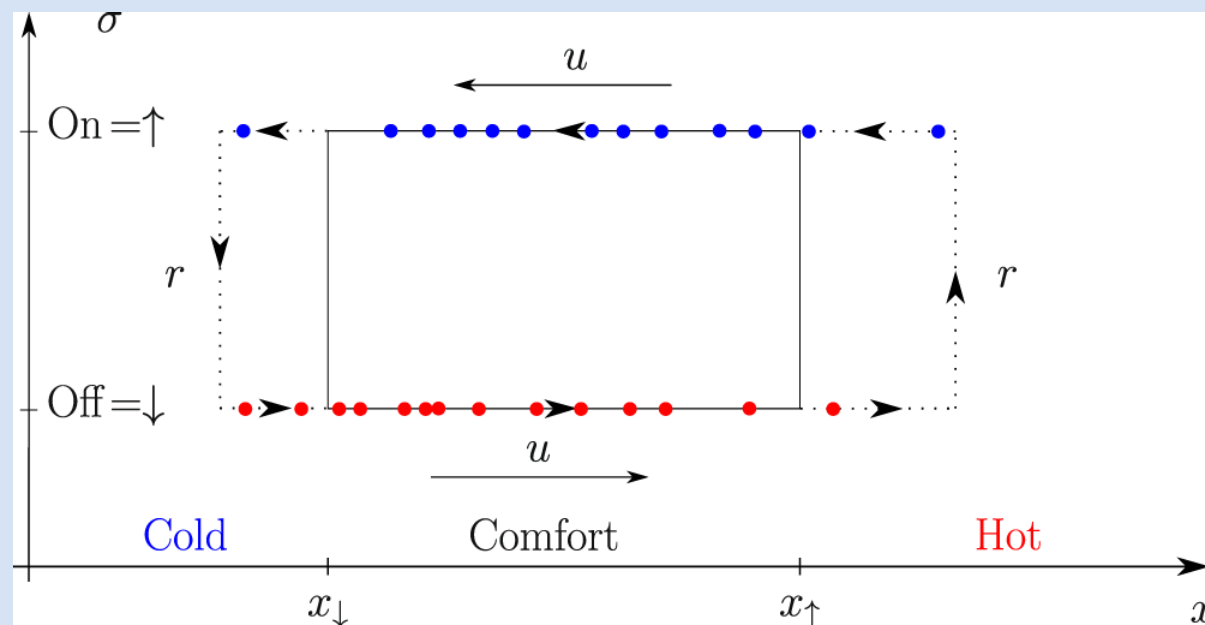
The race follows the theory until it hits the **noise floor** $1/\sqrt{N}$: a finite crowd is never perfectly mixed.

Take-home message

1. **Riemann–Lebesgue:** for $f \in L^1$, $\hat{f}(\xi) \rightarrow 0$. Oscillations cancel against integrable functions, but at no guaranteed rate.
2. **Regularity $\Rightarrow$ decay:** every derivative in L^1 buys a factor $|\xi|^{-1}$; analytic $\Rightarrow$ exponential decay.
3. **Marathon:** free transport moves each speed class as a traveling wave, yet the crowd mixes, with Fourier modes $\hat{g}(kt)$. The smoother the speed distribution, the faster the mixing. This mechanism stabilizes plasma, this is the core of Landau damping!

Fun fact

This mechanism is used in one of my work regarding Power of Ensemble Diversity and Randomization for Energy Aggregation (Métivier et al., 2019, *Scientific Reports*). An ensemble of thermostatically controlled loads e.g. air conditioners turn on/off depending on the temperature. Their distribution actually help mixing them to stabilize the total energy of the ensemble.



Appendix: code

```
1 # The widgets only push their value into an Observable: WGLMakie updates the figure in place.  
2 cancel_law[] = demo_law;
```

▶ 42.5 ms

```
1 cancel_ξ[] = ξ;
```

▶ 2.4 ms

```
1 # The widgets only push their value into an Observable: WGLMakie updates the figure in place.  
2 race_time[] = t_race;
```

▶ 137 ms

```
1 race_data[] = (law=speed_law, v=speeds);
```

▶ 120 ms

```
1 race_show_exact[] = show_exact;
```

▶ 24.8 μs